

VANGUARD AEROSPACE: RMD-03 NEURAL SYNC & INERTIAL DAMPING

Field	Value
Document ID	RMD-03
Classification	TOP SECRET // SCI // NOFORN // ORCON
Owner	Chief Medical Officer
Review Cycle	Quarterly
Revision	6.2.0
Effective Date	2026-05-10
Supersedes	RMD-03 v6.1.2, RMD-03 v6.0.0
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Declassification Review	N/A — Permanent Classification

DIRECTOR'S NOTE: *The pharmacological countermeasures in §2.3 have been significantly updated following the Synaptic Rejection Incident of 2026-04-19, where Pilot Torres experienced Class III rejection during a Class II fold. The previous Neuro-Zeta dosing protocol was insufficient. All flight surgeons must review the revised dosing table before clearing any pilot for fold operations. — Dr. Singh*

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1. CORTICAL SHUNT INTERFACE (CSI)

To operate the TARTARUS-1 craft, the pilot must undergo a surgical implantation of the CSI at the base of the brainstem (medulla oblongata, ventral surface, between the pyramids). The CSI translates human electrochemical thoughts into the craft’s fiber-optic nervous system via a network of **4.2 million quantum-tunneling nanoelectrodes** that interface directly with the pilot’s motor cortex, prefrontal cortex, and cerebellum.

The surgical procedure requires **14 hours** under general anesthesia and a minimum **90-day recovery period**. During recovery, the pilot undergoes progressive neural integration therapy (NIT) to establish baseline coherence between the CSI and the pilot’s neural architecture.

1.1 Sync Coherence Requirements

Coherence Level	Status	Action Required	Maximum Maneuver Authorization	RMD Reference
98% - 100%	Optimal	Full maneuverability enabled	All fold classes (I–III)	RMD-02 §1.1
95% - 97%	Acceptable	Monitor closely; limit to Class I folds if sustained >30 min	Class I–II only	RMD-02 §1.1
90% - 94%	Degraded	Limit to Class I Folds only; notify Chief Medical Officer	Class I only	RMD-02 §1.1
85% - 89%	Critical	Abort mission; initiate auto-hover; prepare for medical extraction	None (hover only)	RMD-05 §3
80% - 84%	Severe	Abort mission immediately; initiate emergency defold per RMD-02 §4	None	RMD-02 §4, RMD-05 §3
< 80%	Rejection	Immediate medical extraction required; initiate Synaptic Rejection Protocol	None	RMD-03 §2.3

IF coherence drops below 90% during a fold operation, **THEN** the QEA enters AUTONAV mode (RMD-02 §3.1). **IF** coherence drops below 80% during a fold, **THEN** the QEA initiates emergency defold and the Chief Medical Officer must be on standby for immediate extraction. **DO NOT** attempt to “push through” degraded coherence — the risk of permanent neural damage is catastrophic.

1.2 CSI-Nanoelectrode Specifications

Parameter	Value	Tolerance	Notes
Total Nanoelectrodes	4.2 million	$\pm 50,000$	Placed during 14-hour surgery
Electrode Material	Gold-Platinum Iridium Alloy	N/A	Biocompatible; resistive degradation < 0.1%/year
Tunneling Current	2.7 nA per electrode	± 0.3 nA	Measured during post-op calibration
Neural Sampling Rate	1,000 Hz	N/A	1,000 neural state samples per second
Data Bandwidth (CSI → QEA)	847 TB/s	N/A	Fiber-optic link to QEA per RMD-02 §3.1
Electrode Lifespan	15 years	± 2 years	Degradation monitoring per §1.3
MRI Compatibility	Conditional	N/A	Must power down CSI before MRI; see §1.4

1.3 Post-Implantation Monitoring Schedule

Time Post-Op	Monitoring Frequency	Tests Required	RMD Reference
0–30 days	Daily	Coherence test, impedance check, cranial MRI	RMD-03 §1.1
30–90 days	Every 3 days	Coherence test, impedance check	RMD-03 §1.1
90–180 days	Weekly	Coherence test, impedance check, cognitive battery	N/A
180–365 days	Bi-weekly	Coherence test, impedance check	N/A
> 365 days	Monthly	Full neural integration panel	N/A

IF electrode impedance increases by >15% from baseline at any monitoring point, **THEN** schedule surgical revision within 30 days. **IF** impedance increases by >30%, **THEN** ground the pilot from all fold operations until revision is complete.

1.4 MRI and Electromagnetic Interference Precautions

The CSI nanoelectrodes are susceptible to damage from strong electromagnetic fields. Pilots with implanted CSIs must observe the following precautions:

Precaution	Requirement	Rationale
MRI Scans	CSI must be powered down 24 hours before scan	RF pulses can desynchronize nanoelectrodes
MRI Strength Limit	No MRI stronger than 1.5 Tesla	Higher field strengths risk electrode migration
Electrosurgery	CSI must be powered down during any electrosurgical procedure	Cautery currents can cascade through nanoelectrode network
Taser/Stun Gun	Absolute prohibition	Even brief exposure causes permanent coherence loss
Cell Phone Proximity	No cell phone within 15 cm of head	Low-power but chronic RF exposure causes drift
Airport Security	Must carry CSI Medical Passport	Standard metal detectors are safe; millimeter-wave scanners are NOT

2. GRAVITON-BASED INERTIAL NULLIFIERS

Because the craft can perform right-angle turns at Mach 20, the biological occupants would experience 40,000 Gs, turning them to plasma. The **Inertial Nullifier** projects a localized graviton field that decouples the cabin's mass from the hull's acceleration. This field is generated by a secondary array of **128 Graviton Emission Pads (GEPs)** distributed throughout the cabin interior.

The Nullifier does not reduce G-forces on the hull — it creates a pocket of “inertially flat” spacetime within the cabin. The pilot and any passengers exist within this pocket, completely shielded from the hull's acceleration.

2.1 Nullifier Calibration

Before any maneuver exceeding 5 Gs, the pilot must mentally “sync” the Nullifier to their specific bone density via the CSI. The Nullifier requires a precise mass model of the pilot's body to calculate the correct graviton emission pattern.

WARNING: If the Nullifier is calibrated to a mass of 80 kg, but the pilot weighs 85 kg, the 5 kg differential will be subjected to the full inertial force of the maneuver, resulting in catastrophic localized shearing of the pilot's extremities. The differential mass is “outside” the inertially flat pocket and experiences raw acceleration.

Calibration Parameter	Measurement Method	Required Accuracy	Consequence of Inaccuracy
Total Body Mass	CSI-integrated bioimpedance scan	± 0.5 kg	Localized shearing (see above)
Bone Density Distribution	Dual-energy X-ray absorptiometry (DEXA)	$\pm 1.0\%$	Bone fracture during high-G maneuvers
Center of Mass	Gravitational pendulum test	± 2.0 mm	Asymmetric inertial protection; rotational shear
Fluid Volume	Bioimpedance analysis	± 50 mL	Internal organ displacement during turns
Spinal Alignment	Cervical/thoracic MRI	± 1.0 mm	CSI electrode migration risk during high-G

IF the pilot's mass changes by >2 kg between calibration and flight (e.g., due to hydration, meals, or injury), **THEN** recalibration is mandatory. **DO NOT** proceed with a maneuver exceeding 5 Gs until recalibration is confirmed by the Chief Medical Officer.

2.2 Nullifier Operational Limits

Parameter	Nominal	Warning	Critical	RMD Reference
Maximum G-Force Compensated	40,000 G	$> 35,000$ G	$> 50,000$ G	RMD-04 §1
Graviton Field Strength	4.72 TeV/m ²	< 4.60 TeV/m ²	< 4.50 TeV/m ²	RMD-01 §1.3
Nullifier Power Draw	210 MW	> 250 MW	> 300 MW	RMD-01 §5
Nullifier Warm-Up Time	30 seconds	N/A	N/A	Must be active before any maneuver >5 G
Maximum Simultaneous Occupants	2	N/A	N/A	3+ occupants exceed graviton field coherence
Field Decay Time (Power Loss)	0.3 seconds	N/A	N/A	Pilot exposed to full G-force within 0.3 s of power loss

IF Nullifier power draw exceeds 250 MW, **THEN** limit maneuvers to <10 G. **IF** power draw exceeds 300 MW, **THEN** initiate COLD IRON per RMD-01 §3 — the Nullifier is failing and any maneuver >2 G will injure the pilot.

2.3 Synaptic Rejection Symptoms and Pharmacological Countermeasures

Prolonged or repeated CSI usage can result in **Synaptic Rejection Syndrome (SRS)**, where the pilot's neural tissue begins to reject the nanoelectrode interface. SRS is classified into four severity classes, each requiring specific pharmacological intervention.

2.3.1 Synaptic Rejection Classification Table

Class	Symptoms	Onset	Coherence Impact	Neuro-Zeta Dose	Adjunct Therapy	RMD Reference
Class I (Mild)	Metallic taste, mild headache, phosphenes (flashing lights), slight tremor in dominant hand	2–6 hours post-flight	Drops to 92–97%	50 mg IV bolus, then 25 mg q6h × 48 hours	Rest, hydration, avoid screens for 24 hours	RMD-03 §1.1
Class II (Moderate)	Epistaxis (nosebleed), temporal dysphoria (time perception distortion), moderate nausea, insomnia, mild memory gaps	1–4 hours post-flight	Drops to 85–91%	100 mg IV bolus, then 50 mg q6h × 72 hours	72-hour grounding (no CSI contact), neurological exam within 24 hours	RMD-03 §1.1, RMD-02 §1.1
Class III (Severe)	Temporary blindness (30–120 s), synaptic misfire (involuntary muscle contractions), tinnitus, severe vertigo, photophobia, transient aphasia	During flight or within 1 hour post-flight	Drops to 80–84%	200 mg IV bolus, then 100 mg q4h × 96 hours	Immediate mission abort; 14-day grounding; full neural panel; surgical review of CSI electrodes	RMD-02 §4, RMD-05 §3

Class	Symptoms	Onset	Coherence Impact	Neuro-Zeta Dose	Adjunct Therapy	RMD Reference
Class IV (Critical)	Cerebral hemorrhage, spontaneous neural cascade failure, loss of consciousness, permanent coherence loss, brain death	During flight	Drops below 80%	400 mg IV bolus (emergency); may repeat once after 30 minutes	Immediate medical extraction; life support; surgical CSI removal within 4 hours or permanent damage	RMD-05 §3, RMD-05 §4

2.3.2 Pharmacological Reference: Neuro-Zeta Inhibitor

Property	Value
Chemical Name	N-(2,6-dimethylphenyl)-2-(tetrazol-1-yl)acetamide
Classification	Selective neural membrane stabilizer
Mechanism of Action	Blocks voltage-gated sodium channels in CSI-adjacent neural tissue; reduces electrochemical interference from nanoelectrode tunneling current
Bioavailability (IV)	100%
Bioavailability (Oral)	62%
Half-Life	4.2 hours
Time to Peak Effect	15 minutes (IV), 45 minutes (oral)
Maximum Safe Daily Dose	400 mg/day
Toxicity Threshold	> 600 mg/day → cardiac arrhythmia risk
Contraindications	history of cardiac arrhythmia, concurrent MAO inhibitors, pregnancy
Side Effects	Drowsiness, mild hepatic enzyme elevation, dry mouth, photosensitivity

2.3.3 Synaptic Rejection Treatment Algorithm

IF pilot exhibits ANY Class I symptoms:

- Administer Neuro-Zeta 50 mg IV
- Monitor coherence for 4 hours
- IF coherence recovers to >95%: clear for limited duty after 24 hours
- IF coherence remains <95%: escalate to Class II protocol

IF pilot exhibits ANY Class II symptoms:

- Administer Neuro-Zeta 100 mg IV
- Ground pilot for 72 hours
- Schedule neurological exam within 24 hours
- IF exam reveals electrode impedance increase >15%: schedule surgical revision
- IF coherence recovers to >97% after 72 hours: clear for Class I maneuvers only

IF pilot exhibits ANY Class III symptoms:

- Administer Neuro-Zeta 200 mg IV
- Ground pilot for minimum 14 days
- Initiate emergency defold per RMD-02 §4
- Full neural panel (EEG, MRI, impedance mapping) within 48 hours
- Surgical review of all 4.2 million nanoelectrodes
- IF coherence does not recover to >90% within 14 days: recommend permanent grounding

IF pilot exhibits ANY Class IV symptoms:

- Administer Neuro-Zeta 400 mg IV (emergency bolus)
- Initiate Protocol Omega per RMD-05 §3
- Surgical CSI removal within 4 hours
- IF brain death confirmed: cease treatment; initiate digital echo upload per RMD-05 §3

DIRECTOR'S NOTE: *The Neuro-Zeta dosing protocol was revised on 2026-05-01 after Pilot Torres (Class III incident, 2026-04-19) required 600 mg over 24 hours — exceeding the previous maximum safe dose. Toxicology analysis showed no cardiac effects, suggesting the toxicity threshold may be higher than 600 mg/day. However, I am not authorizing an increase until Dr. Singh's team completes their follow-up study. — Dr. Vasquez*

3. PILOT ELIGIBILITY AND MEDICAL SCREENING

3.1 Pre-Implantation Requirements

Requirement	Threshold	Test	Frequency
Age	25–40 years	Birth certificate verification	One-time
BMI	18.5–27.5	Standard medical	Pre-implantation
Resting Heart Rate	55–80 bpm	ECG	Pre-implantation
Neural Plasticity Score	> 82/100	Neuropsych battery	Pre-implantation
Bone Density (T-score)	> -1.0	DEXA scan	Pre-implantation
Psychological Stability	Pass Tier-III evaluation	Psychiatric assessment	Pre-implantation, then annually
Genetic Compatibility	HLA match > 85% with CSI alloy	Genetic screening	One-time
No History of Epilepsy	Confirmed via EEG	Electroencephalogram	Pre-implantation
Visual Acuity	20/20 (corrected acceptable)	Standard eye exam	Pre-implantation
Audiometric Threshold	< 25 dB HL at all frequencies	Audiometry	Pre-implantation

3.2 Post-Implantation Grounding Criteria

The pilot must be grounded (prohibited from flight) if ANY of the following conditions are met:

Condition	Grounding Duration	RMD Reference
Coherence < 90% on two consecutive tests	Until coherence recovers to >95% for 72 hours	RMD-03 §1.1
Any Class II or higher rejection episode	Per §2.3.3 treatment algorithm	RMD-03 §2.3
Electrode impedance increase > 15% from baseline	Until surgical revision completed	RMD-03 §1.3
Mass change > 5 kg from Nullifier calibration	Until recalibration confirmed	RMD-03 §2.1
Positive drug screen for CNS depressants	30 days minimum	N/A
Psychological evaluation failure	Until re-evaluation passed	N/A
Unresolved legal or personal stressor	Case-by-case determination	N/A

4. EMERGENCY MEDICAL PROCEDURES

4.1 In-Flight Medical Emergency

If the pilot experiences a medical emergency during flight:

Step	Action	RMD Reference
1	Assess pilot consciousness and coherence level	RMD-03 §1.1
2	IF conscious and coherence >80%: pilot may continue with limitations	RMD-03 §1.1
3	IF conscious and coherence <80%: initiate emergency defold	RMD-02 §4
4	IF unconscious: QEA enters AUTONAV mode	RMD-02 §3.1
5	Notify Base Commander and Chief Medical Officer	RMD-05 §4
6	IF within Class I fold range of base: AUTONAV returns to base	RMD-02 §1.1
7	IF beyond Class I fold range: QEA initiates emergency defold to nearest safe coordinates	RMD-02 §4
8	Ground medical team meets craft upon landing	RMD-05 §4

4.2 CSI Emergency Removal

In extreme cases (Class IV rejection, pilot brain death, or unrecoverable coherence loss), the CSI must be surgically removed. Field removal is possible using the **CSI Emergency Extraction Kit (CSI-EK)**, carried aboard all TARTARUS-1 flights.

CSI-EK Component	Function	Quantity
Cranial access trephine	Opens skull at implant site	1
Nanoelectrode dissolution syringe	Injects gold-platinum solvent to release electrode array	2
Hemostatic gel	Controls bleeding from electrode insertion sites	4
Neural sealant	Prevents cerebrospinal fluid leak post-removal	1
Transport stasis bag	Maintains patient in hypothermic state for transport	1

5. CROSS-REFERENCE INDEX

Reference	Document	Section	Topic
RMD-01 §1.1	Power & Propulsion Interface	Element 115 Decay Chamber	MCF requirements for pilot safety
RMD-01 §1.3	Power & Propulsion Interface	Gravity Wave Amplifiers	GWA power draw and G-force compensation
RMD-01 §3	Power & Propulsion Interface	Emergency Shutdown	COLD IRON procedure
RMD-01 §5	Power & Propulsion Interface	Power Distribution	CSI and Nullifier power priorities
RMD-02 §1.1	Spatial Manifold Navigation	Fold Classes	Biological toll per fold class
RMD-02 §1.2	Spatial Manifold Navigation	Alcubierre-White Tensor	Intent vector parameters
RMD-02 §3.1	Spatial Manifold Navigation	CSI-QEA Integration	Neural intent detection rates
RMD-02 §4	Spatial Manifold Navigation	QEA Diagnostics	Emergency defold procedures
RMD-04 §1	Atmospheric & Trans-Medium Maneuvers	Standard Operating Procedures	G-force compensation during maneuvers
RMD-05 §3	Emergency Containment & Scuttle	Protocol Omega	Fatal scuttle procedure; digital echo upload
RMD-05 §4	Emergency Containment & Scuttle	Evacuation	Medical extraction procedures

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